SENDIT

Project 08 // Airport & Station Item Mail-Home Service
Category: Travel / Smart Infrastructure / Service Engineering
Scope: Global Deployment
Target Interaction Time: < 120 Seconds

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A silent economic and environmental catastrophe repeats itself millions of times a year.

100 Million+

Items confiscated or surrendered globally each year

£12 – £40

Average residual value of an abandoned item

85%

Proportion of items fully re-usable and valuable to their owners



The TSA alone disposes of over 2 million items per month in the United States. To airports, this is waste management. To passengers, it is a direct economic loss.

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The passenger has ninety seconds to decide before the queue breaks down.



Item Flagged

Path 1: Return to Check-in

Action: Leave security, re-queue, pay baggage fee.

Time Cost: 30–60 minutes.

High risk of missing flight.

Path 2: Abandon Item

Action: Drop item into the security bin.

Time Cost: 5 seconds.

Permanent loss of a valuable asset (£60 perfume, £30 knife, £50 power bank).

Travellers treat security screening as a gamble. The loss stems from a single gap: no frictionless option to recover the item.

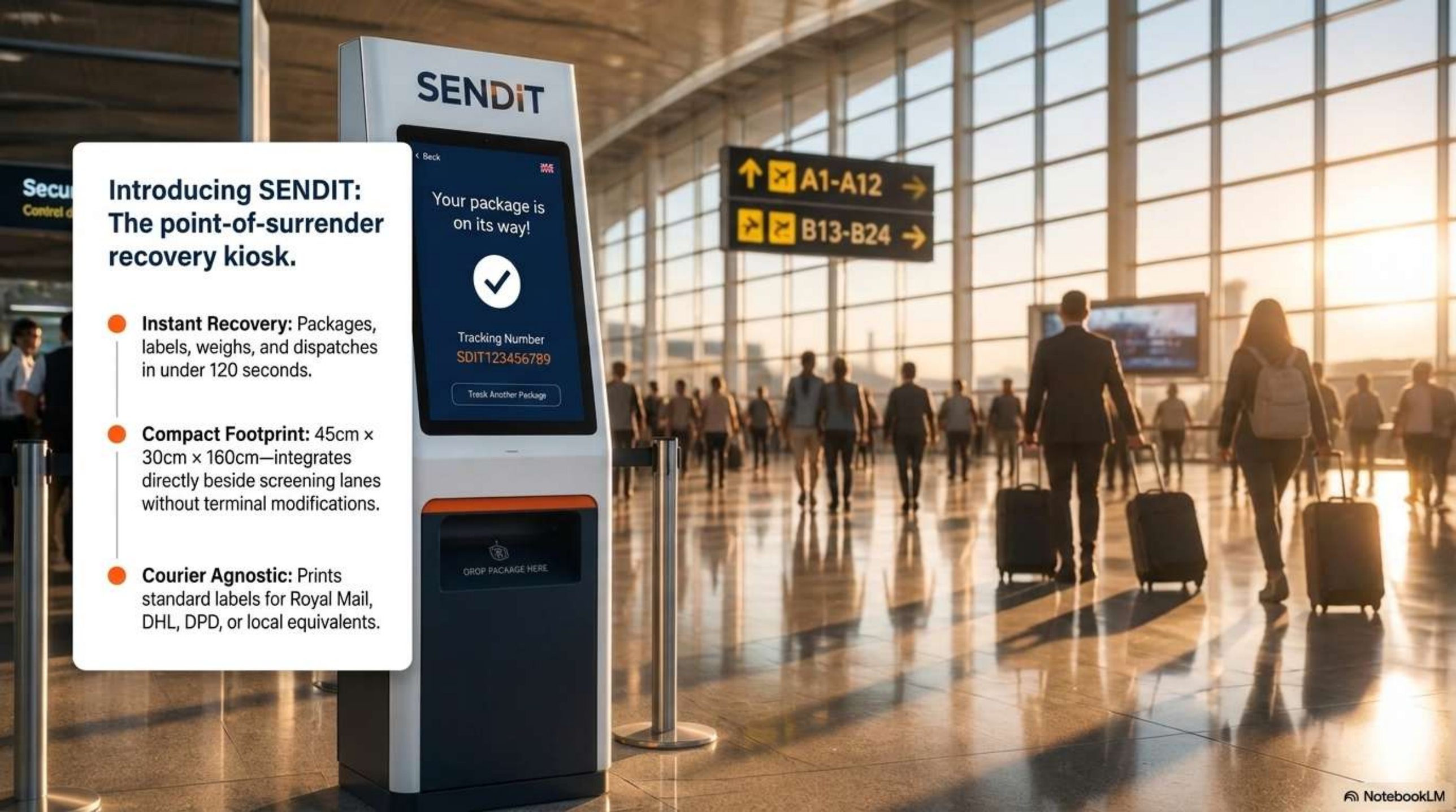
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The market gap is **total:** no standardised, **automated** recovery system exists globally.

	SENDIT (Proposed)	Abandon Item	Airport Hold Counter	Retail Postal Kiosk
Time Required	< 2 mins	5 secs	15-30 mins + return trip	20-40 mins
Passenger Cost	Postage + £3 fee	Total value loss	Storage/Admin fee	Postage + premium box
Airport Capex	Zero - Rev Share	High waste disposal cost	Staffing/Space overhead	N/A
Liability Checking	Automated Edge AI	None	Manual	Self-certified

The only scalable solution must operate entirely within the passenger's sub-2-minute boarding window at the point of surrender.

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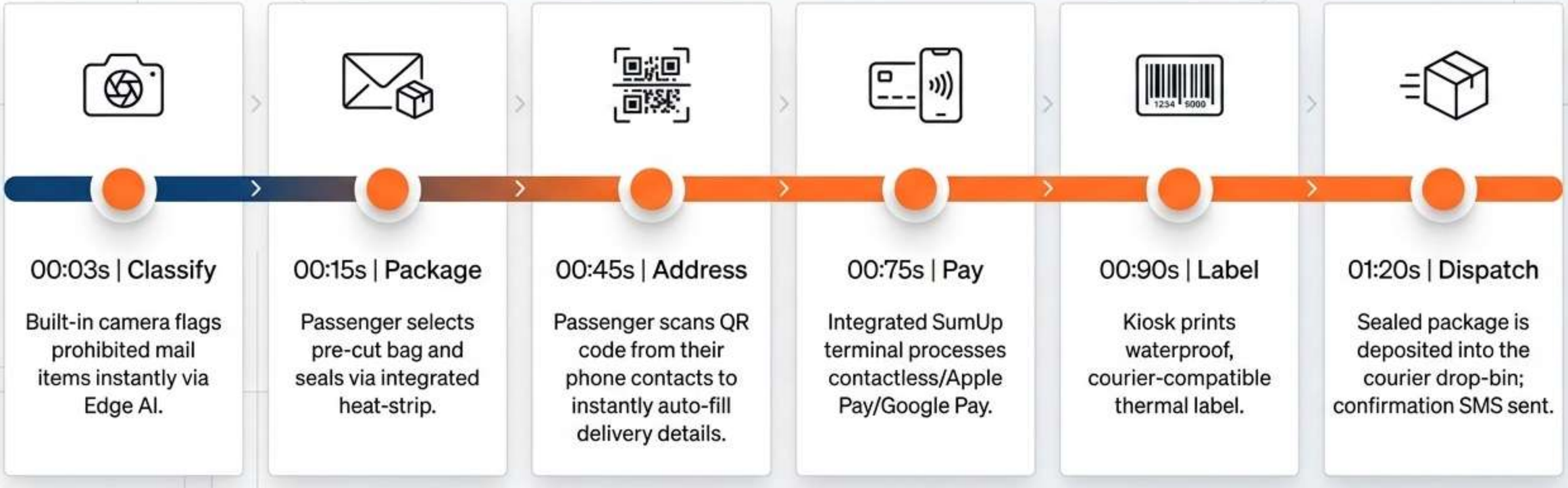


Introducing SENDIT: The point-of-surrender recovery kiosk.

- **Instant Recovery:** Packages, labels, weighs, and dispatches in under 120 seconds.
- **Compact Footprint:** 45cm × 30cm × 160cm—integrates directly beside screening lanes without terminal modifications.
- **Courier Agnostic:** Prints standard labels for Royal Mail, DHL, DPD, or local equivalents.

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The 120-second architecture of seamless asset recovery.



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Engineering out friction: Solving for cognitive load under time pressure.

01 The Address Bottleneck

Observation:
Early user testing revealed passengers struggle to manually type addresses under boarding anxiety.

The Intervention:
A dynamic QR-link system. Passengers scan a QR on the kiosk, tap a saved contact on their phone, and instantly auto-fill the kiosk touchscreen. Zero typing required.

02 The Packaging Bottleneck

Observation:
Sourcing boxes and tape breaks the security lane flow.

The Intervention:
A spring-loaded gravity-fed cassette dispensing three flat-pack bag sizes, combined with a 12V resistive-wire heat-strip sealer built into the shelf.

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Hardware anatomy: Combining consumer-grade edge processing with industrial reliability.

Vision Module

Built-in wide-angle AI Classification Camera.

Payment Gateway

Certified SumUp Pro contactless reader embedded securely behind the bezel.

Enclosure

Injection-moulded ABS with powder-coated steel frame. Total volume BOM: < £3,800.



Printing System

Zebra ZD421 (or DYMO 550) industrial thermal label printer via CUPS driver.

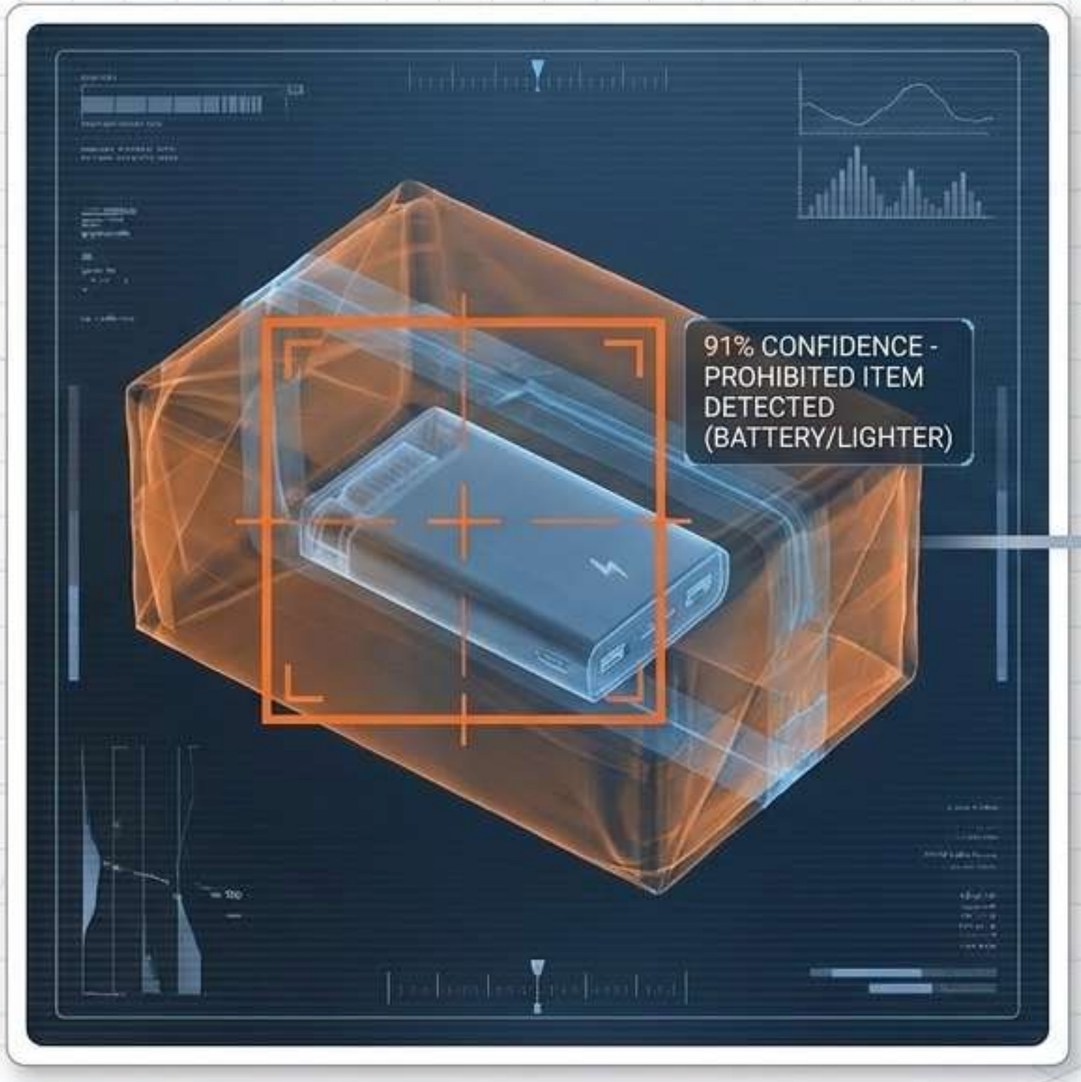
Packaging Hardware

Field-replaceable bag dispenser cassette and relay-controlled 12V heat-strip sealer.

Edge AI Processor

Raspberry Pi 4 (4GB) / NVIDIA Jetson Nano running local Node.js v20 LTS and SQLite (WAL mode).

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Real-time compliance via Edge AI.

Unlike general postal kiosks that rely on self-certification, SENDIT protects courier networks from liability at the point of entry.

Key Specs

Model

MobileNet V3 classification architecture.

Speed

Identifies and cross-references against courier prohibited-mail databases in under 3 seconds.

Training

Initially trained on 12 distinct TSA/IATA prohibited item categories.

Accuracy

91% accuracy achieved in initial prototyping on edge hardware.

Approved for Mail

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Built for **24/7 unattended** operation in high-traffic terminals.

Software & Network

- Dual network connectivity (Wired Ethernet primary + WiFi backup).
- **SQLite** running in Write-Ahead Logging (WAL) mode for high-concurrency database safety.

OS & System Hardening

- Raspberry Pi OS Lite (64-bit) running **Openbox** Kiosk Mode.
- Read-only overlay filesystem protects SD cards.
- **Hardware Watchdog** module triggers automatic reboot if the system hangs for >15 seconds.

Power & Hardware Resilience

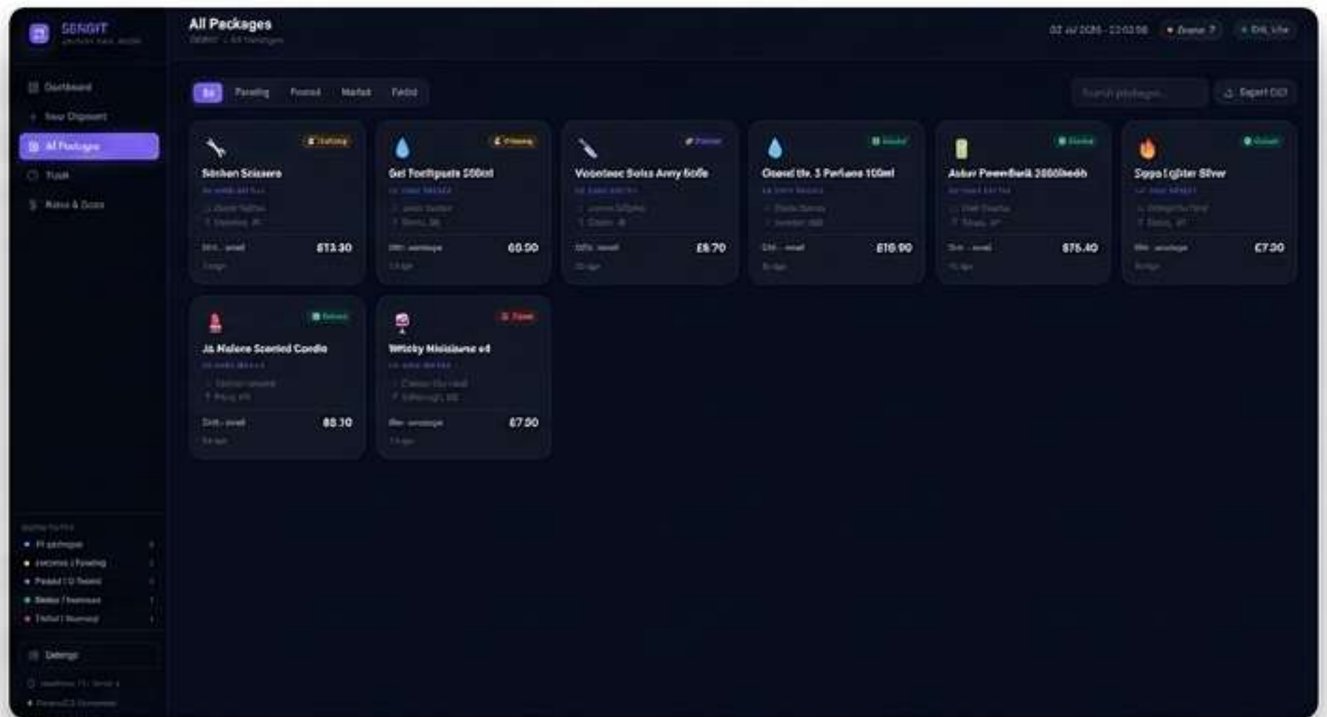
- **Waveshare UPS HAT** prevents database corruption during terminal power cuts.
- **Argon ONE** active cooling prevents thermal throttling above 85°C.

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Bias toward action: The Minimum Viable Prototype.

Proof of Concept Checklist

- **Core Sensing Validated:** Built Pi 5 + camera unit, running local object classification. — Medium
- **Physical Geometry Proven:** 3D-printed bag-sealing shelf and drop-bin housing in PLA. — Medium
- **Financial Flow Live:** Integrated SumUp reader via USB for test transactions. — Low
- **Software Stack Active:** Deployed local Node.js server with SQLite package tracking. — Medium
- **Time Constraint Beaten:** 5-person user testing validated the sub-2-minute interaction claim. — Trivial





Time & Flow

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A **symbiotic ecosystem:** Capturing value from discarded assets.

The Passenger (Asset Recovery)

Converts a total loss into a recovered asset within 1-3 days. Eliminates queue anxiety and travel stress.

The Airport (Revenue & Ops)

Converts a massive waste management overhead into a net-new per-transaction revenue stream. Zero queue disruption.



The Planet (Sustainability)

Diverts millions of high-value manufactured items (electronics, cosmetics, tools) from landfill waste streams into the secondary postal system.

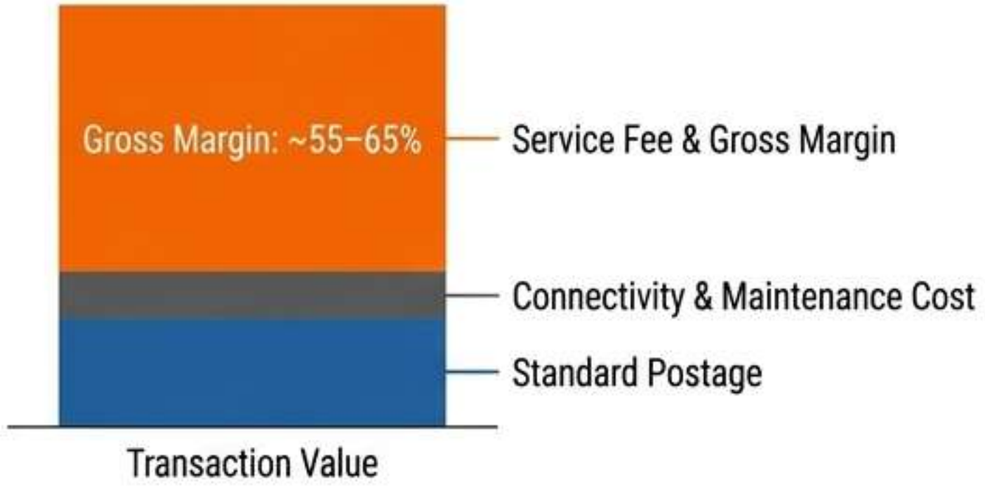
The Courier (Logistics)

Gains a captive, highly concentrated dispatch point for daily high-volume route collection.

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Zero Capex. High Margin. Global Scale.

Unit Economics



- **Hardware** is leased to the airport, not sold. **Zero capital expenditure** for terminal operators.
- **Revenue Model:** Passenger pays standard postage + a £2.50–£4.00 service fee.
- **Gross Margin:** Estimated 55–65% per transaction after connectivity and maintenance.

7.8 Billion

global passengers annually across top 100 airports.

120M – 240M

estimated mailing events annually (at 1.5–3% surrender rate).

£700 Million+

annual global market ceiling.

Time & Flow

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Next steps for deployment and global integration

Milestone 1: The Regional Pilot (Months 1-6)

Deploying at a mid-sized UK regional airport and a Eurostar terminal (where liquid/sharp rules apply).

Target: 500 transactions per terminal/month to validate courier SLAs.

Milestone 2: Top 20 Hubs (Months 6-18)

Scaling the franchised kiosk lease model to the 20 busiest UK and European airports.

Milestone 3: Airline API Integration (Year 2+)

Integrating SENDIT logic into airline apps.

Pre-flag borderline items at home, pre-print a label, and have the bag waiting at a dedicated kiosk bay.

SENDIT. Turning security waste into seamless recovery.

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